

Fractional-Order Dynamics Learning and Control via Data-Driven Approaches: Taking Soft Manipulator as an Example

Xiangyu Shao[✉], *Member, IEEE*, Linke Xu[✉], *Graduate Student Member, IEEE*, Shaojie Zhang[✉],
Guanghui Sun[✉], *Senior Member, IEEE*, and Chengwei Wu[✉]

Abstract—The intrinsic memory and nonlocality that allow fractional-order calculus to capture complex dynamical behaviors also pose significant challenges for accurate modeling and stable control. This article presents a unified data-driven framework that simultaneously addresses these challenges through three key innovations. First, we propose a fractional-order deep Lagrangian network (DeLaN) with a Transformer-like structure, fPLCS-DeLaN, to learn system's inherent fractional-order behaviors directly from uniformly sampled data. It not only enforces fractional-order Lagrangian structure by integrating key physical priors, but also enhances capturing ability of memory effects by incorporating long-short-term convolutional self-attention mechanism. Second, we develop a hybrid network-based disturbance observer, T2F-CRNN, which synergizes CNN's temporal feature extraction, hierarchical recurrence, and interval-based fuzzy inference to robustly estimate uncertainties with unknown nonuniform bounds and capture temporal dependencies. Third, we establish a fully fractional-order controller with practical finite-time convergence. It incorporates input saturation compensation and sliding mode constraints to ensure robustness and high performance. Simulations show that fPLCS-DeLaN achieves modeling errors at least one order of magnitude lower with less than a 15% increase in computational time. The proposed fractional-order controller reduces transient and steady-state tracking errors by 23.1% and 87.6% compared to state-of-the-art controllers, respectively. Experiments on a soft manipulator

platform further demonstrate consistent superiority in model learning and tracking performance.

Index Terms—Deep Lagrangian neural network, fractional-order controller, fractional-order system, soft manipulators, type-2 T-S fuzzy network.

I. INTRODUCTION

A. Background and Literature Review

USING fractional-order calculus to model systems with memory effects has attracted increasing attention in recent years, owing to its superiority in capturing history-dependent, nonlocal, and power-law behaviors [1], [2]. Nevertheless, these features often lead to ill-posed inverse problems [3] and the absence of unified modeling frameworks [4], which challenges accurate model formulation and control strategies design.

These challenges are particularly pronounced in emerging robotic platforms where material memory and geometric nonlinearity coexist. Examples include robotic manipulators with memory-dependent actuation [5] and continuum robots exhibiting fractional-order hysteresis due to viscoelastic material characteristics [6]. Existing modeling approaches can be broadly categorized into physics-based and data-driven paradigms. Classical physics-based methods, including finite-dimensional Lagrangian or Hamiltonian mechanics, offer high fidelity and interpretability but often require detailed knowledge of material parameters and geometric constraints. In contrast, pure data-driven methods treat dynamics as closed-box input-output mappings, lacking physical interpretability and generalizing poorly to unseen conditions.

To bridge the gap between physical fidelity and data-driven flexibility, physics-informed neural networks (PINNs) [7] have enabled data-driven modeling of physical systems by embedding governing equations into learning. Owing to their strong physical interpretability and extrapolation capability, PINNs have been widely applied across diverse domains, ranging from battery systems [8] to biomechanics [9]. To enhance modeling capacity, several advanced PINN variants have been proposed. Fang [10] proposed CNN-based PINNs for spatial operators approximations. Kapoor et al. [11] fused PINN with neural oscillators by exploiting the temporal causality of partial derivative equations (PDEs). Although these methods

Received 23 September 2025; revised 5 January 2026; accepted 12 February 2026. Date of publication 4 March 2026; date of current version 2 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62303137, in part by the Special Funding of China Postdoctoral Science Foundation under Grant 2023TQ0092, in part by the Natural Science Foundation of Sichuan Province under Grant 2025ZNSFSC0461, in part by the New Era Heilongjiang Outstanding Master's and Doctoral Dissertation Foundation under Grant LJYXL2023-029, in part by Heilongjiang Postdoctoral Science Foundation under Grant LBH-Z20057, in part by the Pre-Research Task of State Key Laboratory of Robotics and System [Harbin Institute of Technology (HIT)] under Grant SKLRS202404B, and in part by the Ph.D. Candidate Innovation Fund of State Key Laboratory of Robotics and Systems under Grant SKLRS-2025-ZM-10. (Corresponding author: Guanghui Sun.)

Xiangyu Shao, Guanghui Sun, and Chengwei Wu are with the State Key Laboratory of Robotics and System (HIT) and the School of Astronautics, Harbin Institute of Technology, Harbin 150001, China, and also with Suzhou Research Institute of HIT, Suzhou 215104, China (e-mail: xiangyushao@hit.edu.cn; guanghuisun@hit.edu.cn; chengweiwu@hit.edu.cn).

Linke Xu and Shaojie Zhang are with the State Key Laboratory of Robotics and System (HIT) and the School of Astronautics, Harbin Institute of Technology, Harbin 150001, China (e-mail: linkexu@stu.hit.edu.cn; sj_zhang@stu.hit.edu.cn).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TNNLS.2026.3665811>, provided by the authors.

Digital Object Identifier 10.1109/TNNLS.2026.3665811

TABLE I
COMPARISON OF PHYSICS-INFORMED MODELING APPROACHES

Method	Nonloc. Supp.	Var. Struct.	Temp. Aware.	Uni-Spa. Data Comp.	Dyn. Mod. App.	Real Ver.
Standard PINNs [7]	✗	✗	✗	✗	▲	▲
CNN-PINNs [10]	✗	✗	✓	✗	✗	✗
Neural Oscillator PINNs [11]	✗	✗	✓	✗	▲	✗
fPINNs [13]	✓	✗	✓	✗	▲	▲
Score-fPINNs [14]	✓	✗	✓	✗	✗	✗
MC-tfPINNs [15]	✓	✗	✓	✗	▲	✗
FFNN [16]	✓	✗	✓	✗	✗	✗
DeLaN [17]	✗	✓	✗	✓	✓	✓
ServoLNN [18]	✗	✓	✗	✓	✓	✗
Self-Attentive DeLaN [19]	✗	✓	▲	✓	✓	✓
Ours (fPLCS-DeLaN)	✓✓	✓	✓	✓	✓	✓

✓✓: Supports simultaneous learning of nonlocality and the fractional orders; ✓: Yes/Supports; ▲: Partially/Only for weakened versions; ✗: No/Does not support.

improve time-series modeling and generalization, they fail to capture nonlocality and power-law behaviors or enforce structural physical constraints. From a control-oriented perspective, Hillebrecht and Unger [12] derived a posteriori error bounds for PDE-defined PINNs, enabling certified modeling in feedback design. Yet their framework relies on semigroup-theoretic assumptions that fundamentally conflict with the nonlocal and non-Markovian nature of fractional-order dynamics, limiting its applicability.

As for modeling long-range temporal dependencies, fractional-order PINNs (fPINNs) incorporate nonlocal fractional derivatives into the physics-informed framework. Pioneered by Pang et al. [13], fPINNs have enabled data-driven solution and parameter identification of fractional-order PDEs, with recent variants improving numerical stability through Monte Carlo integration, score-based formulations, and analytically tractable basis functions [14], [15], [16]. Despite these advances, existing fPINNs treat system dynamics as unstructured input–output mappings. These methods neither enforce fundamental variational principles essential for energy-based modeling, nor embed explicit mechanisms to encode memory-dependent physical laws, rendering them unsuitable for model-based control of mechanical systems and compromising their extrapolation capabilities.

This limitation is particularly critical for mechanical systems, where Lagrangian formulations are broadly used for energy-based dynamics modeling, even in the presence of non-conservative or history-dependent effects. To bridge this gap, structure-preserving models like the Deep Lagrangian Network (DeLaN) [17] embed the Euler–Lagrange structure directly into the network architecture. However, most applications treat DeLaN as a plug-and-play module, with few architectural innovations that fundamentally enhance its expressivity. Notable exceptions include ServoLNN [18], which adapts to varying servo inputs, and an attention-enhanced variant [19]. Nevertheless, to the best of our knowledge, no existing work extends DeLaN to systems with long-term memory, nor significantly modifies its architecture to capture their memory effects.

While accurate modeling of history-dependent systems is essential, its ultimate purpose lies in enabling reliable control, especially in the presence of uncertainties and other practical

constraints. Existing robust strategies generally fall into two categories: 1) observer-based approaches, which reconstruct lumped uncertainties by disturbance observers like sliding mode observers [20], [21] and recurrent neural network-based observers [22], [23], [24] and 2) direct adaptive compensation schemes, which compensate uncertainties through adaptive control laws or robust terms without explicit observer design [25], [26], [27]. Despite their effectiveness, both paradigms commonly assume that the uncertainty remains within a constant bound. This assumption becomes restrictive when dealing with history-dependent uncertainties. Moreover, these approaches have not been adapted to fractional-order systems, where model uncertainties may exhibit long-range temporal dependence that cannot be accurately captured by existing disturbance observers.

B. Contributions

This article addresses the aforementioned challenges by establishing a fractional-order Probabilistic Sparse Long-Short-Term Convolutional Self-Attention DeLaN (fPLCS-DeLaN), which constitutes one of the earliest efforts that explicitly embed memory effects into a variational learning architecture. While recent fPINN variants have advanced numerical stability through basis-function representations [16] or have been extended to Lévy-driven stochastic PDEs via fractional score functions [14], they remain confined to specific PDE classes without enforcing embedded variational principles. Moreover, their reliance on dense and arbitrarily sampled collocation points is incompatible with the fixed-frequency data acquisition mechanism in practical systems. To bridge this gap, fPLCS-DeLaN integrates a long-short-term convolutional self-attention mechanism that dynamically weights historical states according to their relevance to current dynamics, enabling accurate modeling of temporal dependence from sparse and uniformly sampled data. Table I compares the proposed approach with existing methods across nonlocality support, variational structure, temporal awareness, sparse and uniform data compatibility, complex system applicability, and real-world verification.

While fPLCS-DeLaN provides a structured and memory-aware modeling approach, its real-world deployment neces-

sitates enhanced robustness against unmodeled dynamics and external disturbances. In most practical applications, accurate control relies on compensation of lumped uncertainties. However, in complex memory-dependent systems, these uncertainties often exhibit long-range temporal correlations and unknown nonuniform bounds, challenging existing RNN-based observers [22], [23], [24]. To address this, we then introduce Type-2 Fuzzy Convolutional Recurrent Neural Network (T2F-CRNN), a hybrid disturbance observer that combines CNN-RNN for spatiotemporal feature learning with Type-2 T-S fuzzy systems for interval-based, interpretable uncertainty modeling without requiring uniform bounds. Furthermore, we develop a fully fractional-order control framework that simultaneously handles modeling uncertainties, input saturation, and sliding mode constraints. In addition, the proposed controller achieves practical finite-time convergence, ensuring robust trajectory tracking and aligning with the memory effects of fractional-order systems.

The main contributions are as follows.

- 1) Propose fPLCS-DeLaN, a structure-preserving deep architecture tailored for modeling dynamics with long-range temporal dependence. It effectively captures the long-range memory-dependent behaviors by integrating long-short-term convolutional attention mechanism into Transformer-like network structure.
- 2) Develop a T2F-CRNN disturbance observer, which synergizes CNN for temporal-dependent feature extraction, hierarchical recurrence for state evolution modeling, and interval Type-2 T-S fuzzy inference for uncertainty estimation. This observer can effectively model the memory effects of lumped uncertainties in fractional-order systems.
- 3) Establish a fully fractional-order controller based on the learned dynamics. It enforces practical finite-time stability and prescribed performance in the presence of input saturation and uncertainties with unknown nonuniform bounds.
- 4) Present comprehensive simulations and real-world experiments on soft manipulators, showing consistent superiority over state-of-the-art baselines in tracking accuracy and robustness both theoretically and practically.

The remainder of this article is organized as follows. Section II presents the proposed fPLCS-DeLaN for fractional-order dynamics modeling. Section III describes the design of the fractional-order controller integrated with the network-based disturbance observer. Section IV reports the simulation results. Section V further validates the approach with real-world experiments conducted on a soft manipulator. Finally, Section VI concludes this article.

II. FRACTIONAL-ORDER DYNAMICS LEARNING VIA DATA-DRIVEN APPROACHES

Traditional DeLaN incorporates Lagrangian mechanics into neural networks to enhance the model's extrapolation and physical interpretability. However, it typically relies on single-step error minimization, which is insufficient for capturing

long-range dependencies. To model systems with long-range temporal dependence, this section develops a fractional-order deep Euler-Lagrangian network with Transformer-like architecture, referred to as fPLCS-DeLaN.

A. Preliminaries

Definition 1: [28]: The α -order Caputo derivative is

$${}_CD_t^\alpha f(t) = \begin{cases} \frac{d^m f(t)}{dt^m}, & \alpha = m \\ \frac{1}{\Gamma(m-\alpha)} \int_0^t \frac{D^m f(\tau)}{(t-\tau)^{\alpha-m+1}} d\tau, & m-1 < \alpha < m \end{cases}$$

where $\alpha \in \mathbb{R}^+$, $m \in \mathbb{N}^+$, and $\Gamma(z) = \int_0^{+\infty} t^{z-1} e^{-t} dt$ is the Gamma function. ${}_CD_x^\alpha$ is abbreviated as D_x^α , and D^α is fractional derivatives with respect to time. For positive nonintegers α and ϵ , the α -order ϵ -Caputo calculus is

$$\epsilon D_t^\alpha f(t) = \frac{1}{\Gamma(m-\alpha)} \int_{\max(t-\epsilon, 0)}^t \frac{D^m f(\tau)}{(t-\tau)^{\alpha-m+1}} d\tau$$

in which m is an integer that satisfies $m-1 < \alpha < m$.

B. Fractional-Order Lagrangian Dynamics and Its Data-Driven Modeling Framework

According to [17], the traditional n -dimensional integer-order Lagrangian dynamics can be formulated as

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \tau \quad (1)$$

in which $L = T - V$ is the Lagrangian function with $q, \tau \in \mathbb{R}^n$ denoting the generalized coordinates and control inputs, respectively. The kinetic energy $T = 1/2 \dot{q}^T B(q) \dot{q}$, where the inertia matrix $B(q) \in \mathbb{R}^{n \times n} > 0$ is symmetric and positive definite. From (1), one has

$$\frac{d}{dt} (B(q) \dot{q}) - \frac{1}{2} \frac{\partial}{\partial q} (\dot{q}^T B(q) \dot{q}) + \frac{\partial V(q)}{\partial q} = \tau. \quad (2)$$

Following [5], (2) is reformulated as:

$$D^\beta (B(q) D^\alpha q) - \frac{1}{2} \frac{\partial}{\partial q} ((D^\alpha q)^T B(q) D^\alpha q) + \frac{\partial V(q)}{\partial q} + N(q) D^\alpha q = A(q) \tau \quad (3)$$

where $0 < \alpha, \beta < 1$ are unknown fractional orders, $N(q), A(q) \in \mathbb{R}^{n \times n} > 0$ denote the damping and input matrices. To ensure positive definiteness of $B(q)$, $N(q)$, we parameterize them using the Cholesky decomposition

$$\begin{aligned} B(q) &= L_B(q) L_B^T(q) + \epsilon_a I_n \\ N(q) &= L_N(q) L_N^T(q) + \epsilon_n I_n \end{aligned} \quad (4)$$

where $\epsilon_a > 0$ is a small regularization constant to avoid singularities. This parameterization reduces the number of learnable parameters to $(n(n+1))/2$ per matrix while preserving $B(q)$, $N(q) > 0$. To ensure the positivity of the diagonal values, L_B and L_N are designed as

$$\begin{aligned} L_B &= \text{diag}(\sigma(\hat{\xi}_{B1} + \epsilon_b \mathbf{1}_n)) + \text{SLT}(\hat{\xi}_{B2}) \\ L_N &= \text{diag}(\sigma(\hat{\xi}_{N1} + \epsilon_b \mathbf{1}_n)) + \text{SLT}(\hat{\xi}_{N2}) \end{aligned} \quad (5)$$

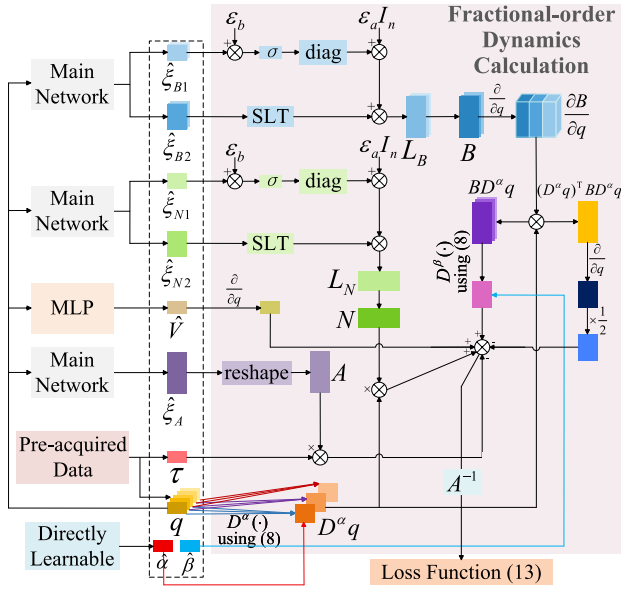


Fig. 1. Pipeline of fPLCS-DeLaN for fractional-order dynamics calculation.

where $\hat{\xi}_{B1}, \hat{\xi}_{N1} \in \mathbb{R}^n$, $\hat{\xi}_{B2}, \hat{\xi}_{N2} \in \mathbb{R}^{(n(n-1))/2}$ are the network outputs. $\sigma(\cdot)$ is the Sigmoid function and $\text{SLT}(\cdot)$ rearranges a vector of length $(n(n-1))/2$ into a n -order strictly lower triangular matrix by filling its subdiagonal entries row-wise. The input transform matrix $A(q)$ is obtained by reshaping another network output $\hat{\xi}_A \in \mathbb{R}^{n^2}$ into an $n \times n$ matrix in row-major order.

Since the Caputo fractional derivative in (3) depend on the entire continuous history of the state trajectory, it cannot be directly evaluated from discrete-time measurements, which are typical in practice. Consequently, a discrete approximation incorporating a finite history of past states is necessary for data-driven learning. Although a truncated Riemann–Liouville formulation with finite memory horizon was proposed in [29], the Caputo definition remains preferable for physical modeling due to its clear interpretation of initial conditions, particularly those with direct mechanical significance, making it suitable for a wide range of engineering applications. Adopting the high-order time-stepping discretization scheme with cubic interpolation proposed by Kumari et al. [30] and taking $\zeta(t) = t$, $\omega_n = 1$, the Caputo fractional-order derivative is approximately discretized as

$$D^\alpha f_N = \sum_{i=0}^{L_1} \lambda_i f_{N-i}, \quad \lambda_i = \sum_{j=1}^{L_2+1} A_{j,i+1-j} \quad (6)$$

in which $L_1 > L_2 \in \mathbb{N}^+$ represent the number of involved historical steps and the interpolation order, respectively. $A_{j,i}$ is computed by L_2 -order interpolation, given by

$$\Pi_i^{L_2} f(t) = \sum_{r=0}^{L_2} f_{l-r} \prod_{s=0, s \neq r}^{L_2} \left(\frac{t - t_{l-s}}{t_{l-r} - t_{l-s}} \right) \quad (7)$$

where t_i denotes the i th uniform sample with step size Δt . Differentiating w.r.t. t yields

$$\frac{d \Pi_i^{L_2} f(t)}{dt} = \sum_{r=0}^{L_2} f_{l-r} M_{r,i}^{L_2}(t) \quad (8)$$

in which $M_{r,i}^{L_2}(t)$ is a $(L_2 - 1)$ -order polynomial, $\sum_{i=1}^{L_2+1} A_{i,k} f_{l+1-i}$ is obtained by the following integral:

$$\frac{1}{\Gamma(1-\alpha)} \int_{t_{l-1}}^{t_l} [t_{k+l} - t_s]^{-\alpha} \frac{d \Pi_s^{L_2} f(t)}{dt} ds. \quad (9)$$

The coefficient $A_{i,k} = A_{i,k}(\alpha)$ depends solely on the order α . Substituting (4)–(6) into (3), we obtain a closed-form representation of the fractional-order Lagrangian dynamics parameterized by the network outputs

$$\hat{\tau} = \text{Dyn}(\hat{\xi}_{B1}, \hat{\xi}_{B2}, \hat{\xi}_{N1}, \hat{\xi}_{N2}, \hat{\xi}_A, \hat{V}, \hat{\alpha}, \hat{\beta}; q, q^{[-i]}) \quad (10)$$

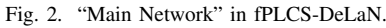
in which $q^{[-i]}$ denotes states from i previous steps, $i = 1, 2, \dots, 2L_1$. The loss function is defined as

$$F_{\text{loss}} = \text{mean}_{q, \tau} [\|\hat{\tau} - \tau\|^2] + \epsilon_w L_w \quad (11)$$

where the “mean” is taken over all training samples of q and τ , $\epsilon_w > 0$, L_w denotes the weight regularization term. Fig. 1 illustrates the computational pipeline of fPLCS-DeLaN, showing how the network outputs are mapped to the fractional-order dynamics and physical loss evaluation. The vectors $\hat{\xi}_{B1}$, $\hat{\xi}_{B2}$, $\hat{\xi}_{N1}$, $\hat{\xi}_{N2}$, and $\hat{\xi}_A$ are learned by three dedicated subnetworks, which together constitute the core of fPLCS-DeLaN. The potential energy $V(q)$ is parameterized via a MLP, denoted as $\hat{V}(q)$, to maintain computational efficiency. The fractional orders α and β are learnable unknown constants and estimated as $\hat{\alpha}$, $\hat{\beta}$. In the diagram, overlapped variable modules indicate temporal sequences: three instances represent a history of L_1 past steps (total length $L_1 + 1$), while five instances represent $2L_1$ past steps (total length $2L_1 + 1$).

Remark 1: The high-order cubic interpolation scheme from [30] is adopted to balance accuracy and computational tractability, while allowing flexible selection of the interpolation order. Traditional lower-order methods, such as Grünwald–Letnikov or L1 schemes, incur significant truncation errors when processing long-term memory. High-accuracy alternatives like full-history adaptive quadrature or spectral methods introduce prohibitive computational overhead, and often lack the fixed-length history window and smooth differentiability required for gradient-based learning. Frequency-domain methods, like Oustaloup filters, are inherently linear and thus unable to represent state-dependent nonlinear dynamics within a unified modeling framework. In contrast, this approach provides differentiable, adjustable-memory discretization with higher-order fidelity, making it well-suited for gradient-based learning of models governed by long-range memory effects.

In practice, many complex fractional-order systems exhibit strong nonlinearity and long-term memory effects, making them difficult to model or predict. There is a pressing need for a neural network capable of effectively learning from limited observational data while incorporating physical constraints and capturing the long-term dependencies. To tackle this challenge, we propose fPLCS-DeLaN, a novel Transformer-inspired extension of DeLaN with long-short-term convolutional attention mechanism. The “Main network” of fPLCS-DeLaN (shown in Fig. 2) consists of three major components: input layer, encoder layer, and decoder layer. The



Part 1: Input Layer: Sequential system configurations $q^{[-i]}$, $i = 0, 1, \dots, L_1$ sampled at fixed intervals Δt are preprocessed through cleaning, filtering, vectorization, and temporal alignment. The resulting multistep sequences that incorporate both current and historical states are critical for capturing unknown memory effects. Before being fed into the encoder, the sequence is mapped through an embedding layer that combines token and sinusoidal positional embeddings via element-wise summation, yielding the positional encoding formulation for each element

$$PE_{2i} = \sin\left(\frac{n_{pos}}{\exp\left(\frac{16i}{d_m}\right)}\right), PE_{2i+1} = \cos\left(\frac{n_{pos}}{\exp\left(\frac{16i}{d_m}\right)}\right) \quad (12)$$

Part 2: Encoder Layer: Inspired by the Informer architecture [31], the proposed encoder adopts a pyramidal architecture composed of stacked hierarchical layers. Each layer contains a Multihead ProbSparse Convolutional Self-attention module and an Add & Norm module. The multihead self-attention mechanism projects the input into multiple subspaces, where attention weights are independently computed. This allows the model to capture long-range dependencies and diverse patterns in parallel. Each head computes its own key (k), query (q), and value (v), derives attention scores, and contributes to the concatenated output. By combining multiple attention heads,

Depicted in the bottom-left of Fig. 2, the ProbSparse LST-Convolutional self-attention layer includes five modules: parallel long-short-term convolutional (LST-Conv) modules, a concatenation-and-projection layer, a sparsity measurement layer, a query sparsification layer, and a final attention score computation layer (not explicitly shown as it follows the standard attention mechanism). Each LST-Conv module integrates dilated causal convolution for long-term dependencies with standard convolution for short-term features in a parallel fashion. By stacking multiple dilated causal convolution layers with exponentially increasing dilation rates, the receptive field expands exponentially with depth, enabling the model to capture long-range temporal dependencies without excessive parameter growth. The dilated causal convolution operates on the input sequence with a dilation factor d , the output at a given time step is

$$y = \sum_{k=0}^{K-1} w_k x^{[-dk]} \quad (13)$$

In sparsity measurement layer, attention sparsity is quantified by the divergence of the attention distribution induced by query q_i over all keys k

$$\begin{aligned} \mathcal{S}(q_i, k) = & \log \left(\sum_{j=1}^{\text{len}(k)} K(q_i | k_j) \right) - \log(\text{len}(k)) \\ & - \frac{1}{\text{len}(k)} \sum_{j=1}^{\text{len}(k)} \log(K(q_i | k_j)) \end{aligned} \quad (14)$$

in which distinct attention scores $K(q_i k_j) = \exp((q_i k_j^T) / (\sqrt{d}))$, q_i and k_j are the query and key components, d is the scaling factor determined by the embedding dimension. High sparsity scores reflect concentration in the attention weights. Accordingly, the sparsification layer retains only the top- u entries to reduce complexity with $u = c_q \ln(\ln(q))$, in which c_q is a constant sampling factor. Consequently, the feature dimension decreases progressively across layers, forming a pyramidal structure that aggregates attention and sparsity features across all layers.

Part 3: Decoder Layer: The decoder receives two inputs: the encoder feature map and a sequence generated by simulating a stochastic differential equation

$$dx_t = \mu(x_t, t) dt + \sigma(x_t, t) dW_t \quad (15)$$

where $\mu(x_t, t)$ is the deterministic drift, $\sigma(x_t, t)$ is the diffusion coefficient (random fluctuation intensity), and W_t is a standard Brownian motion. This introduces controller variability to enhance robustness and generalization.

The decoder comprises:

- 1) **Masked Multihead Probsparse LST-Convolutional Self-Attention Layers** Within each attention head, the input is either the preprocessed time series data or the preceding layer outputs. This input is processed through parallel LST-Conv modules, and their outputs are linearly projected to obtain query, key, and value representations. The queries are then sparsified via the ProbSparse mechanism, retaining only the top- u entries with highest sparsity scores [computed via (14)]. The attention output is computed as $\text{Attn}(q, k, v) = \text{softmax}((qk^T) / (\sqrt{d_k}))v$ with a causal mask applied to the attention scores before the softmax operation to ensure autoregressive property. Finally, the outputs from all heads are concatenated to produce the final output of the attention module. This layer is the same as Multihead LST-Convolutional Probsparse Self-attention layer in the encoder, except for a mask operation after LST-Conv modules.
- 2) **Multihead Attention Layers.** The query and key vectors are derived from the encoder outputs, and the value vectors are obtained from the preceding layer in the decoder.
- 3) **An Attention Score Calculation Layer.**
- 4) **An output MLP layer.** The decoder outputs flattened vectors representing the function to be fit, whose values can be traced to the current state q and historical motion states $q^{[-i]}$. Similar to the original Transformer framework, the decoder layer can be stacked to form a deep network that progressively refines the output.

Remark 2: Although the learned vectors, $(\hat{\xi}_{B1}, \hat{\xi}_{B2}, \hat{\xi}_{N1}, \hat{\xi}_{N2}, \hat{\xi}_A)$, depend only on the current state q , real-world dynamics rarely satisfy (3) exactly due to hysteresis, viscoelasticity and sensor noise. To account for these nonideal effects, fPLCS-DeLaN incorporates a history window of past states $q^{[-i]}$, $i = 0, 1, \dots, L_1$. This temporal augmentation allows the network to capture long-term dependencies intrinsic to fractional-order dynamics while enhancing robustness and generalization.

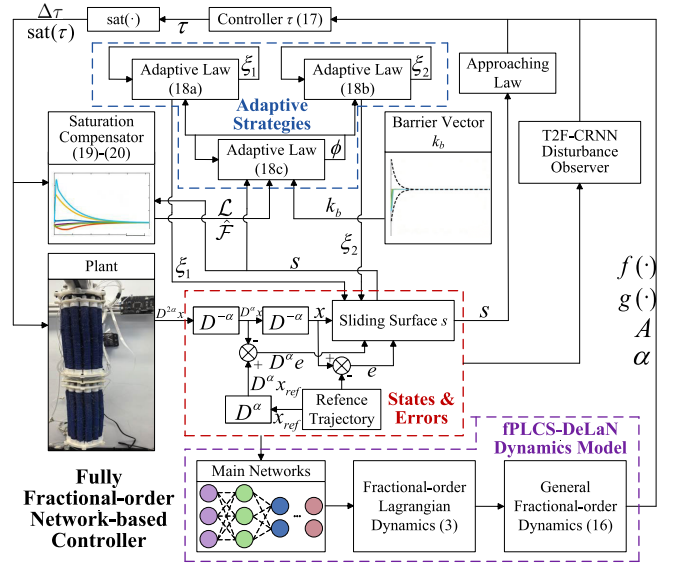


Fig. 3. Overview of the fully fractional-order controller. It comprises a model learned by fractional-order Lagrangian network, a T2F-CRNN disturbance observer, input saturation compensator, sliding mode constraints, and parameter adaptive strategies.

III. FRACTIONAL-ORDER CONTROL DESIGN WITH NETWORK-BASED DISTURBANCE OBSERVER

This section presents the fully fractional-order controller. Considering the complexity and unknown bounds of the system uncertainties, T2F-CRNN is proposed to estimate the unknown lumped uncertainties. An overview of the proposed control architecture is illustrated in Fig. 3.

A. Fractional-Order Controller

The fractional-order dynamics can be expressed as

$$\begin{aligned} D^\alpha x_1 &= f(x_1, x_2) \\ D^\alpha x_2 &= \text{Asat}(\tau) + \mathcal{F} + g(x_1, x_2) \end{aligned} \quad (16)$$

where $x_1, x_2 \in \mathbb{R}^n$ are system states, $\tau \in \mathbb{R}^n$ is the control input with its upper bound vector $\tau_m \in \mathbb{R}^n > \mathbf{0}_n$, $\text{sat}(\tau) \in [-\tau_m, \tau_m]$ denotes the saturation function. $\mathcal{F} = \mathcal{F}_{\text{model}} + \mathcal{F}_{\text{ext}} \in \mathbb{R}^n$ represents the lumped uncertainties arising from modeling error and external disturbance. The fractional order $0 < \alpha < 1$, $f(x_1, x_2)$, $g(x_1, x_2)$ are both differentiable. The input transform matrix $A \in \mathbb{R}^{n \times n}$ is invertible, differentiable, and bounded. For simplicity, we assume a common fractional order $\alpha \in (0, 1)$ for both state equations, where $\alpha = (\alpha_{\text{net}} + \beta_{\text{net}})/2$ and α_{net} , β_{net} are predicted by the fractional-order Lagrangian network.

The controller is designed as

$$\begin{aligned} \tau &= -A^{-1} \left(\hat{\mathcal{F}} + g(x_1, x_2) - D^\alpha (x_{2\text{ref}} + \xi_2) \right. \\ &\quad + c(f(x_1, x_2) - D^\alpha (x_{1\text{ref}} + \xi_1)) \\ &\quad + \mathcal{P}_1(x_1, x_2) \text{sgn}(s) - \frac{D^\alpha k_b^2}{2} \otimes \text{invx}(s) \\ &\quad \left. + k_1 s + k_2 \text{sig}^\gamma s \right) + k_3 \phi \end{aligned} \quad (17)$$

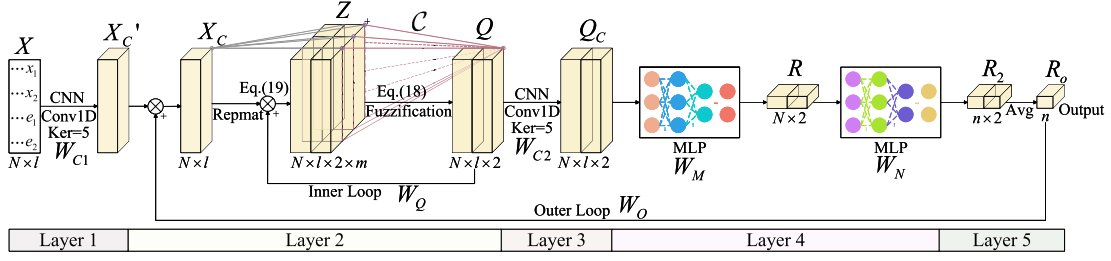


Fig. 4. Structure diagram of T2F-CRNN.

with the following adaptive laws:

$$D^\alpha \xi_1 = -k_4 \xi_1 - k_7 \text{sig}^\gamma \xi_1 - \mathcal{P}_2(s, \xi_1) \text{sgn}(\xi_1) + \Lambda \xi_2 + \phi \quad (18a)$$

$$D^\alpha \xi_2 = -k_5 \xi_2 - k_8 \text{sig}^\gamma \xi_2 - \mathcal{P}_3(s, \xi_2) \text{sgn}(\xi_2) + \phi \quad (18b)$$

$$D^\alpha \phi = -k_6 \phi \otimes (k_b^2 - s^2) - k_9 \text{sig}^\gamma \phi + \Delta \tau - \mathcal{L} \quad (18c)$$

where $k_1, \dots, k_6, \Lambda > 0$, the sliding surface $s = e_2 + ce_1$, $e_1 = x_1 - x_{1ref} - \xi_1$, $e_2 = x_2 - x_{2ref} - \xi_2$, x_{1ref} and x_{2ref} are the reference of x_1 and x_2 . $k_b = k_b(t) > 0_n$ is a monotonically decreasing element-wise barrier vector that constrain s within range $(-k_b, k_b)$. $\mathcal{P}_i(\cdot)$ are undetermined positive definite matrix functions. Symbols $\otimes, \oslash$ represent the Hadamard product and division. $\text{invx}(x) = 1/x$ when $|x| > \varepsilon$ and $\text{invx}(x) = 1/(\varepsilon \text{sgn}(x))$ when $|x| \leq \varepsilon$. $\text{sig}^\gamma(x) = |x|^\gamma \text{sgn}(x)$. Both $\text{invx}(\cdot)$ and $\text{sig}(\cdot)$ are element-wise. The saturation function $\text{satx}(\cdot)$ is defined as

$$\text{satx}(x) = \begin{cases} x, & |x| < \tau_m - \epsilon \\ \frac{4\epsilon\tau_m - (\tau_m + \epsilon - x)^2}{4\epsilon\text{sgn}(x)}, & \tau_m - \epsilon \leq |x| \leq \tau_m + \epsilon \\ \tau_m \text{sgn}(x), & |x| > \tau_m + \epsilon \end{cases} \quad (19)$$

with $0 < \epsilon \ll \tau_m$, and τ_m is the bounds of $\text{satx}(\cdot)$. The elements of $\mathcal{L}$ are defined as

$$\mathcal{L}_i = \begin{cases} \frac{\frac{r_1 \Delta \tau_i^2}{k_{bi}^2 - s_i^2} + r_2 \text{sig}^\gamma(\Delta \tau_i) + r_3 s_i^2}{\epsilon_\phi}, & |\phi_i| \geq \epsilon_\phi \\ \frac{\frac{r_1 \Delta \tau_i^2}{k_{bi}^2 - s_i^2} + r_2 \text{sig}^\gamma(\Delta \tau_i) + r_3 s_i^2}{\epsilon_\phi}, & |\phi_i| < \epsilon_\phi \end{cases} \quad (20)$$

in which $\epsilon_\phi > 0$ is a small value, $r_1, r_2, r_3 > 0$, $\Delta \tau = \text{satx}(\tau) - \tau$. $\hat{\mathcal{F}}$ is the estimated lumped uncertainties output by disturbance observer, which will be defined in Section III-B.

B. Network-Based Disturbance Observer

As shown in Fig. 4, the T2F-CRNN used for the estimation of lumped uncertainty consists of five layers: CNN input layer, fuzzification layer with fully connected internal recurrence, CNN intermediate layer, multilayer perceptron layer, and output layer with external recurrence.

Layer 1: CNN Input Layer: The network input is constructed by concatenating x_1, x_2, e_1, e_2 , and the saturated input $\text{sat}(\tau)$ along both channel and temporal dimensions with a fixed historical length, forming $X \in \mathbb{R}^{N \times l}$, where $N = 5n$ is the channel number and l the time length. The input sequence is

processed by 1-D CNN using a kernel W_{C1} of size 5 with the same padding, producing X'_C with a shape identical to X . Then X'_C is fused with the external recurrent component X_R through element-wise addition, yielding the final layer output $X_C = X'_C + X_R$.

Layer 2: Fuzzification Layer With Fully Connected Internal Recurrence: In the fuzzification layer, each element of X_C is mapped through the Type-2 T-S fuzzification process. Specifically, $2m$ learnable centers are defined for the membership functions, denoted as $\mathcal{C} = [c_{11}, c_{12}, c_{21}, c_{22}, \dots, c_{m1}, c_{m2}]$, where $c_{i1} < c_{i2}$. For each element $x_c \in X_C$, the lower and upper Type-2 membership functions, $\underline{\mu}_i$ and $\bar{\mu}_i$, are then computed as

$$\begin{aligned} \underline{\mu}_i &= \begin{cases} \mathcal{G}_\sigma(\underline{z}_i, c_{i2}), & x_c < \frac{c_{i1} + c_{i2}}{2} \\ \mathcal{G}_\sigma(\bar{z}_i, c_{i1}), & x_c \geq \frac{c_{i1} + c_{i2}}{2} \end{cases} \\ \bar{\mu}_i &= \begin{cases} \mathcal{G}_\sigma(\bar{z}_i, c_{i1}), & x_c < c_{i1} \\ 1, & c_{i1} \leq x_c \leq c_{i2} \\ \mathcal{G}_\sigma(\underline{z}_i, c_{i2}), & x_c > c_{i2}. \end{cases} \end{aligned} \quad (21)$$

Here, $\mathcal{G}_\sigma(z, c) = \exp(-(z - c)^2 / (2\sigma^2))$ is the Gaussian function. The recurrent states $\underline{z}_i$ and $\bar{z}_i$ are computed from the input element x_c by (24) and stored in a 4-D tensor $Z \in \mathbb{R}^{N \times l \times 2 \times m}$. Define basis functions as

$$\varphi_i(x) = \sin(\Omega_i x + \theta_i) \quad (22)$$

where Ω_i and θ_i are predefined. Then, the output corresponding to each element of X_C is expressed as

$$\underline{q} = \sum_{i=1}^m \underline{\mu}_i \varphi_i(\underline{z}_i), \quad \bar{q} = \sum_{i=1}^m \bar{\mu}_i \varphi_i(\bar{z}_i) \quad (23)$$

in which Q is $N \times l \times 2$. The recurrent weight W_Q has a shape of $N \times l \times 2 \times N \times m$. For each $x_c \in X_C$, recurrent connections are established with all elements of Q at the same time step, yielding $2N$ links. Each link is associated with a weight vector of length m , resulting in $2Nm$ recurrent weights for x_c , denoted as w_{qijk} with indices $i = 1, \dots, N$, $j = 1, \dots, m$, and $k \in \{1, 2\}$, where $k = 1$ and $k = 2$ correspond to $\underline{q}$ and $\bar{q}$, respectively. The recurrent sums $\underline{z}_j$ and $\bar{z}_j$ are then computed as

$$\underline{z}_j = x_c + \sum_{i=1}^N w_{qij1} \mathcal{Z}(\underline{q}_i), \quad \bar{z}_j = x_c + \sum_{i=1}^N w_{qij2} \mathcal{Z}(\bar{q}_i) \quad (24)$$

where q_i and $\bar{q}_i$ denote the i th lower and upper bounds of Q at the given time step. The operator $\mathcal{Z}(\cdot)$ refers to the value propagated from the previous time step.

Layer 3: CNN Intermediate Layer: This CNN layer performs temporal transformation on the time series data. The kernel W_{C2} has a shape of 5×2 , and operates independently on $\underline{Q}$, $\bar{Q}$. Using the same padding strategy, the output Q_C preserves the shape of Q , while maintaining separate lower and upper bounds.

Layer 4: Multilayer Perceptron: This layer reduces the temporal dimension while extracting both linear and nonlinear features from time series. The layer output R is denoted as

$$R = W_{Md} \circ \sigma \circ W_{M(d-1)} \circ \sigma \circ \dots \circ W_{M1} \circ \sigma(Q_C) \quad (25)$$

in which $W_{Mi} \in \mathbb{R}^{w_i \times w_{i-1}}$, $i = 1, \dots, d$ denote the weight matrices of the network layers, d is the network depth, w_i the width of i th layer, with $w_0 = l$, $w_d = 1$. All matrix multiplications are performed along the temporal dimension, and σ is the activation function. The output R is $N \times 2$.

Layer 5: Output Layer With External Recurrence: The expected output is an n -dimensional vector. To achieve this, the layer includes a shallow MLP that transforms the channel dimension from N to n . The final output R_o is then obtained by averaging the two bounds. Let $R_2 = [r^{(1)}, r^{(2)}] \in \mathbb{R}^{n \times 2}$, where $r^{(1)}$ and $r^{(2)}$ denote the lower and upper bound vectors, respectively. Then this layer can be expressed as

$$R_o = \frac{r^{(1)} + r^{(2)}}{2} \in \mathbb{R}^n \quad (26)$$

where the multilayer perceptron includes learnable weights W_{Ni} , $i = 1, \dots, v$. The final output is then fed back to the input layer through l weight matrices of shape $N \times n$, denoted as $W_o \in \mathbb{R}^{N \times n \times l}$. The recurrent input is given by $X_R = \mathcal{W}_o(R_o) \in \mathbb{R}^{N \times l}$, where $\mathcal{W}_o$ applies l learned linear projections $W_o^{(i)} \in \mathbb{R}^{N \times n}$ to R_o across the temporal dimension.

In T2F-CRNN, the estimated lumped uncertainties are represented by $\hat{F} = R_o$. The learnable parameters include $W_{C1}, C, W_Q, W_{C2}, W_{Mi}, W_{Nj}, W_o$, where $i = 1, \dots, d$, $j = 1, \dots, v$. These learnable parameters are grouped into seven vectors p_i , $i = 1, 2, \dots, 7$. The update law for these parameters can be written as

$$D^\alpha p_i = \eta_i J_i s \odot (k_b^2 - s^2) - Z_i p_i - S_i \text{sig}^\rho p_i - Q_i(s, p_i) p_i \quad (27)$$

where $J_i = (\partial R_o) / (\partial p_i)$ is the Jacobian matrix, $Z_i, S_i, \eta_i > 0$, $0 < \rho < 1$, each $Q_i(\cdot)$ is an undetermined positive definite function. The practical finite-time stability proof of the proposed controller is detailed in Section I of the Supplementary Materials.

IV. SIMULATION RESULTS

A. FPLCS-DeLaN Modeling Method

A pneumatically actuated soft manipulator is applied in the simulations to verify the effectiveness of the proposed method. According to [21], the integer-order dynamics of a two-segment soft manipulator can be written as

$$B(q) \ddot{q} + C(q, \dot{q}) \dot{q} + K(q) q + N(q) \dot{q} + G(q) = A(q) \tau \quad (28)$$

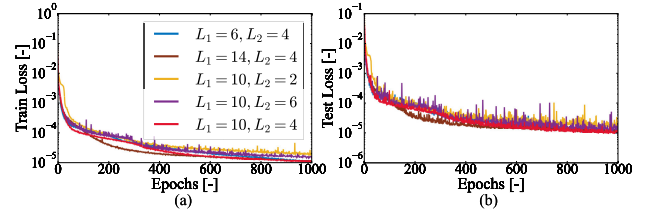


Fig. 5. (a) Training and (b) testing loss for different L_1 and L_2 in fPLCS-DeLaN.

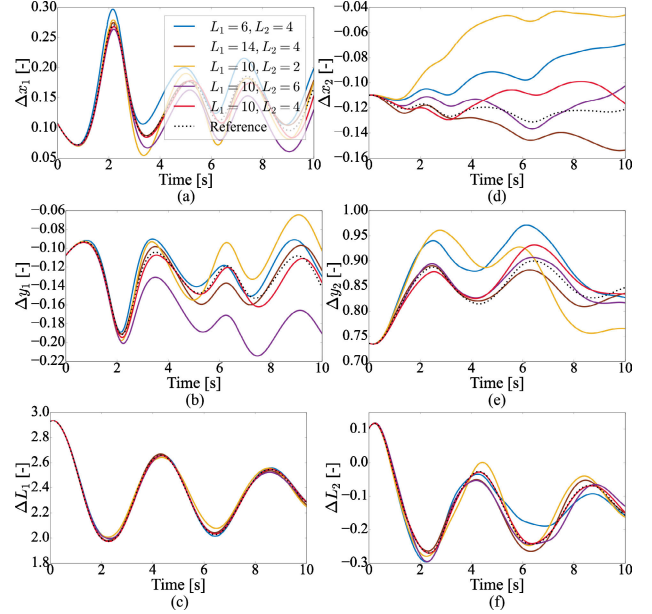


Fig. 6. 10-s open-loop-state prediction for different L_1 and L_2 in fPLCS-DeLaN. (a)–(f) Δx_1 , Δy_1 , ΔL_1 , Δx_2 , Δy_2 , and ΔL_2 .

where $q = [\Delta x_1, \Delta y_1, \Delta L_1, \Delta x_2, \Delta y_2, \Delta L_2]^T \in \mathbb{R}^6$, $\tau = [\tau_{11}, \tau_{12}, \tau_{13}, \tau_{21}, \tau_{22}, \tau_{23}]^T \in \mathbb{R}^6$ are generalized coordinates and control inputs, $B(q), C(q, \dot{q}), K(q), N(q), A(q) \in \mathbb{R}^{6 \times 6}$ represent the inertia, Coriolis, stiffness, damping and input transform matrices, respectively. $G(q) \in \mathbb{R}^6$ is the gravity term. Here, Δx_i , Δy_i , and ΔL_i denote the Δ -parameterization of Segment i of the soft manipulator, representing its planar displacement and length variation, respectively. The control inputs τ_{j1} , τ_{j2} , and τ_{j3} correspond to two bending torques and one axial force applied at the distal end of Segment j , following [21]. According to (3), the fractional-order dynamics for the soft manipulator has the following form:

$$D^\beta (B(q) D^\alpha q) + \bar{C}(q, D^\alpha q) D^\alpha q + K(q) q + N(q) D^\alpha q + G(q) = A(q) \tau \quad (29)$$

where $\alpha, \beta \in (0, 1)$ are unknown fractional orders and $\bar{C}(q, D^\alpha q) = -1/2 \partial / (\partial q) (B(q) D^\alpha q)$.

In simulations for fPLCS-DeLaN, we first perform parameter sensitivity analyses on the number of historical steps L_1 and the interpolation order L_2 , followed by ablation and comparative studies. The configurations, dataset and hyperparameters are detailed in Section II of the Supplementary Materials through Fig. S1 and Table S1.

1) Parameter Sensitivity Study: This study evaluates the performance of fPLCS-DeLaN under different L_1 and L_2 . The

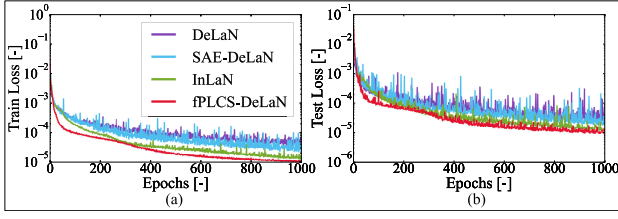


Fig. 7. (a) Training and (b) testing loss for different methods.

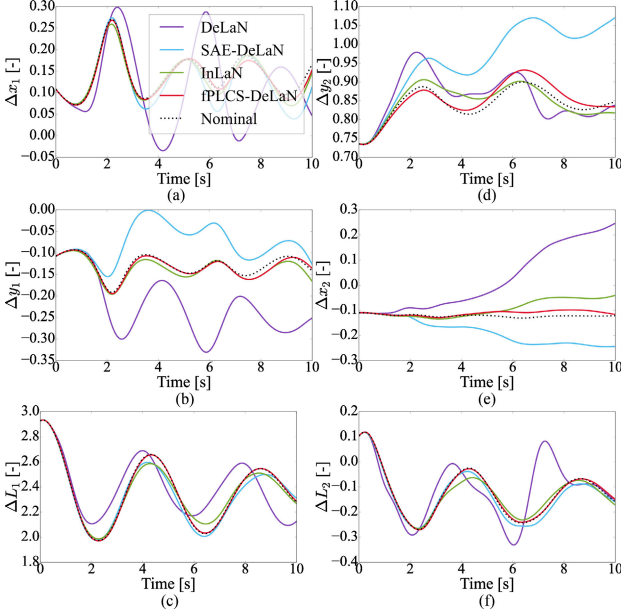


Fig. 8. 10-s open-loop state prediction for different methods. (a)–(f) Δx_1 , Δy_1 , ΔL_1 , Δx_2 , Δy_2 , and ΔL_2 .

loss curves and 10-s open-loop trajectories (see [19], [32] for details) are presented in Figs. 5 and 6.

From Fig. 5, the loss curves for fPLCS-DeLaN with $L_1 = 10$, $L_2 = 4$ remain the most stable. Compared with $L_1 = 6$, $L_2 = 4$ and $L_1 = 10$, $L_2 = 2$, this configuration also achieves a lower final loss, reflecting improved approximation accuracy. This advantage is further validated by the open-loop performance in Fig. 6, where $L_1 = 10$, $L_2 = 4$ yields the highest extrapolation and prediction accuracy. Small L_1 or L_2 is prone to underfitting and inaccurate estimation. Conversely, large L_1 may cause overfitting, higher computational cost, and degraded attention focus. Large L_2 introduces numerical instabilities that result in unstable training and higher prediction errors.

2) *Ablation and Comparative Study*: This part compares fPLCS-DeLaN with existing DeLaNs such as SAE-DeLaN [19] and traditional DeLaN [17]. The setting $L_1 = 10$, $L_2 = 4$ is adopted across all control groups. For ablation studies, we introduce InLaN, a variant that removes the LST-Conv modules and adopts an Informer-style decoder [31]. The loss curves and 10-s open-loop performance of the four DeLaN architectures are shown in Figs. 7 and 8.

Fig. 7 shows that fPLCS-DeLaN achieves the most stable training convergence among all compared models. Its testing loss curves show improved generalization and long-term memory acquisition, owing to the integration of LST-Conv layers and a Transformer-like architecture. Fig. 8 further

TABLE II
MSE AND TRAINING TIME FOR DIFFERENT L_1, L_2 , AND DeLaN ARCHITECTURES

Parameter/Network	MSE	Time Cost (s)
$L_1 = 6, L_2 = 4$	4.962×10^{-3}	15385
$L_1 = 14, L_2 = 4$	8.579×10^{-4}	16431
$L_1 = 10, L_2 = 2$	9.654×10^{-3}	15328
$L_1 = 10, L_2 = 6$	2.942×10^{-3}	16670
$L_1 = 10, L_2 = 4$	6.393×10^{-4}	15859
InLaN	4.353×10^{-3}	15771
SAE-DeLaN	3.104×10^{-2}	13827
DeLaN	9.514×10^{-2}	11416

validates its superior accuracy in state prediction, as the LST-Conv layers effectively capture the memory effects and enable high-accuracy long-term forecasting. In contrast, InLaN and SAE-DeLaN exhibit degraded performance due to their limitation in modeling long-range temporal dependencies. Quantitative results are summarized in Table II, where the first five rows evaluate fPLCS-DeLaN under different L_1 and L_2 , the last three rows benchmark existing models and the ablation variant.

According to Table II, the proposed fPLCS-DeLaN achieves a favorable tradeoff between accuracy and efficiency. It attains higher accuracy than existing baselines with only a modest increase in training time. Specifically, among all parameter settings, the configuration $L_1 = 10$, $L_2 = 4$ yields the lowest mean squared error (mse) while maintaining a moderate training cost. Decreasing L_1 to 6 or L_2 to 2 saves only 2.99% and 3.35% in training time, respectively, but at the cost of a 6.8-fold and 14.1-fold increase in mse, clearly indicating diminishing returns. Compared to InLaN, SAE-DeLaN, and DeLaN, fPLCS-DeLaN requires only 0.56%, 14.7%, and 38.9% more training time, yet achieves a mse reduced by factors of 6.81, 48.6, and 149, respectively. This demonstrates that the additional computational cost is well justified by the significant gains in modeling accuracy.

B. Fractional-Order Controller and Network-Based Observer

This section evaluates the proposed fractional-order controller and the T2F-CRNN disturbance observer, based on the proposed fractional-order dynamics of a two-segment soft manipulator (29) with order $\alpha = \beta = 0.5$. Sinusoidal external disturbance $\mathcal{F}_{\text{ext}}$ is added on each dimension with a magnitude of 2 and 0.5 Hz frequency. The hyperparameters of the controller are provided in Table S3 of the Supplementary Materials. The simulation results are shown in Fig. 9.

As shown in Fig. 9, the proposed method achieves fast and accurate trajectory tracking, with the sliding surface maintained within the prescribed region. The control input initially reaches the limits but quickly decreases to a reasonable range. The estimation error becomes stable after 2.5 s. Furthermore, ξ_1 , ξ_2 , and ϕ converge to the origin and remain bounded within the expected limits.

1) *Comparison of T2F-CRNN With Other Existing Network-Based Observers*: To further verify the effectiveness of the proposed T2F-CRNN disturbance observer, we compare

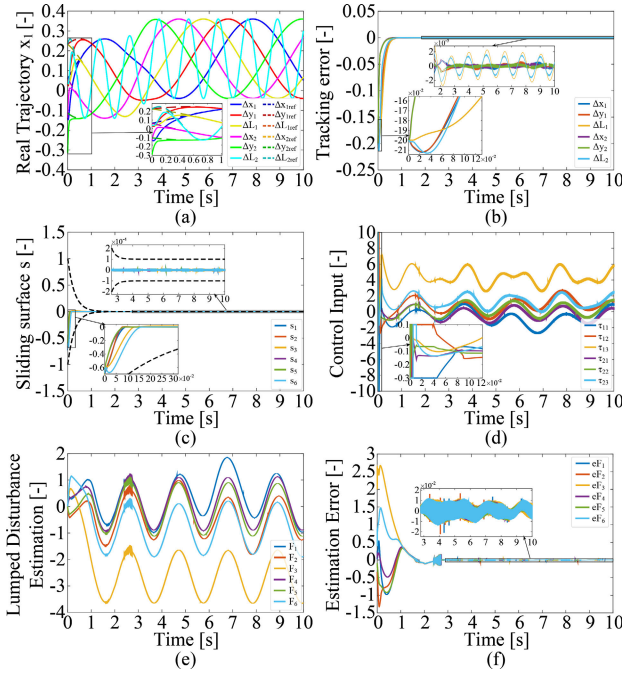


Fig. 9. Simulation results under the proposed controller and observer. (a) Real states x_1 (solid lines) and the reference x_{1ref} (dashed lines) of the soft manipulator. (b) Tracking error $x_1 - x_{1ref}$. (c) Sliding surface s . (d) Saturated control input $\text{sat}x(\tau)$. (e) Estimated lumped uncertainties $\hat{F}$ by T2F-CRNN. (f) Estimation error $\hat{F} - \tilde{F}$.

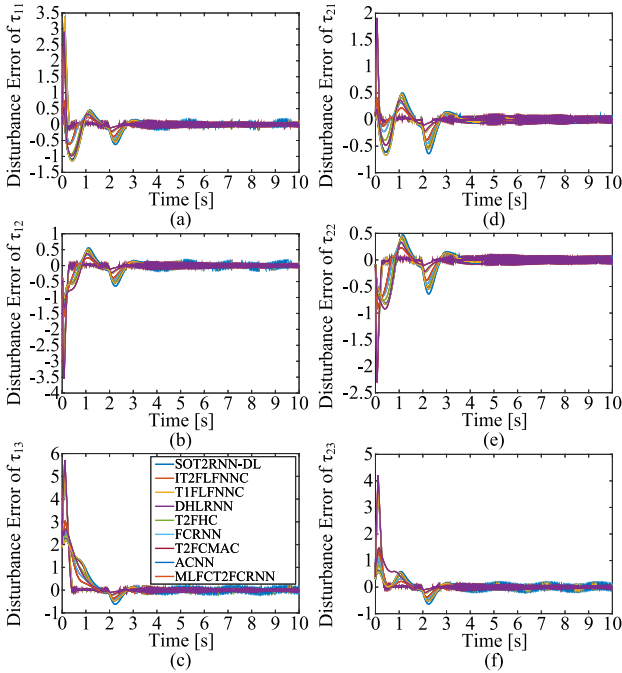


Fig. 10. Uncertainty estimation errors of different observers. (a)–(f) τ_{11} , τ_{12} , τ_{13} , τ_{21} , τ_{22} , and τ_{23} .

the proposed T2F-CRNN observer with SOT2RNN-DL [24], IT2FLFNNC [23], T1FLFNNC [23], DHLRNN [22], T2FHC [33], FCRNN [34], T2FCMAC [35], and actor-critic neural network (ACNN) [36]. The T2F-CRNN is configured with a history window of $l = 7$ steps, $2m = 82$ radial basis function centers, and width parameter $\sigma = 5$. The first MLP has depth $d = 5$ with 30-unit hidden layers, and the second has depth

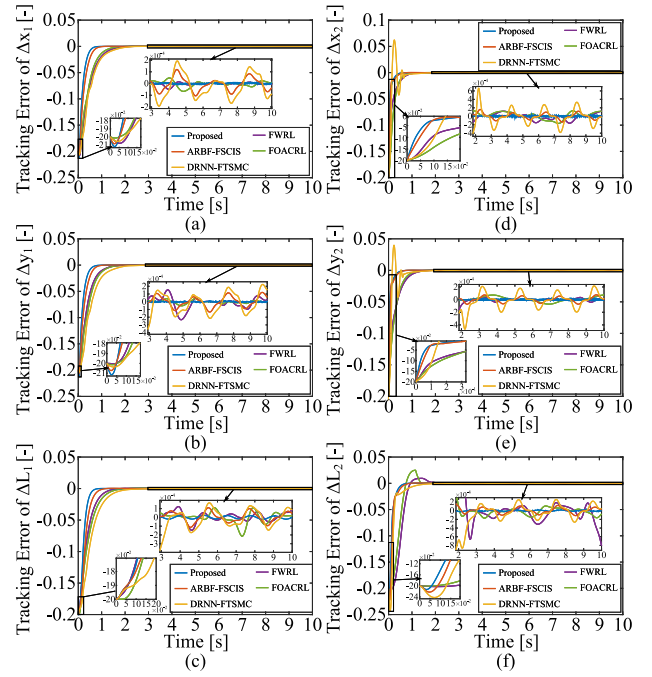


Fig. 11. Tracking error under the five controllers. (a)–(f) Δx_1 , Δy_1 , ΔL_1 , Δx_2 , Δy_2 , and ΔL_2 .

TABLE III

METE FOR THE FIVE CONTROLLERS IN CONVERGING AND STEADY STAGE. THE BEST RESULTS ARE IN BOLD

Controller	METE at [0, 2]s	METE at [5, 10]s
ARBF-FSCIS	5.4534×10^{-2}	1.3047×10^{-4}
DRNN-FTSMC	1.0532×10^{-1}	3.1659×10^{-4}
FWRL	9.9695×10^{-2}	1.9604×10^{-4}
FOACRL	9.7988×10^{-2}	1.3389×10^{-4}
Our controller	4.1920×10^{-2}	1.6101×10^{-5}

$v = 3$ with 20-unit hidden layers. The basis functions $\varphi_i(x)$ employ frequency parameters $\Omega_i = 0.35i$ and phase shifts θ_i randomly sampled from $[-\pi, \pi]$. Fig. 10 presents the tracking and disturbance estimation error under the aforementioned network-based observers.

As shown in Fig. 10, T2F-CRNN exhibits smaller initial overshoot in disturbance estimation. Once the system reaches steady state at $t \approx 3$ s, the estimation error remains close to zero, whereas DHLRNN and ACNN exhibit pronounced oscillations.

2) *Controller Performance Comparison*: This section compares the proposed controller with four representative baseline controllers for fractional-order systems: adaptive RBF NN tracking controller with full-state constraints and input saturation (ARBF-FSCIS) [37], deep RNN with finite-time terminal sliding mode controller (DRNN-FTSMC) [38], fuzzy weight-based reinforcement learning (FWRL) [39] and fractional-order optimal controller via reinforcement learning (FOACRL) [40]. The hyperparameters of these controllers are detailed in Table S3 of the Supplementary Materials. The tracking errors under these controllers are shown in Fig. 11.

Fig. 11 shows that the proposed controller achieves faster convergence speed and lower steady-state error, with negli-

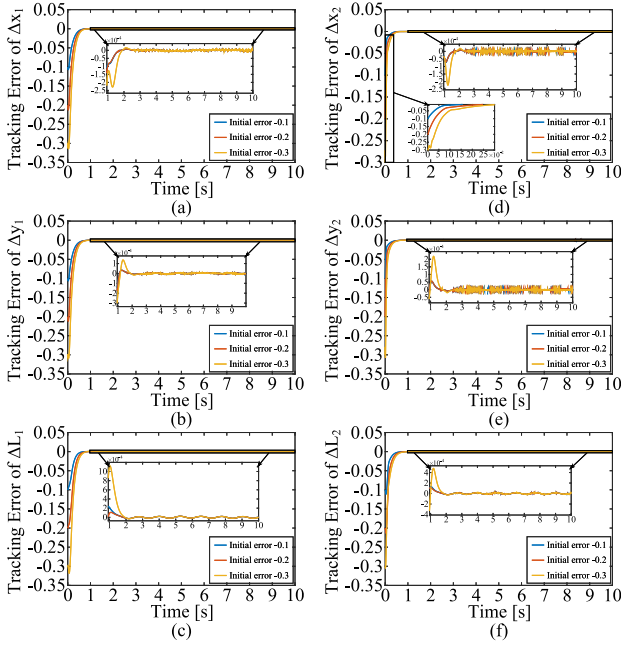


Fig. 12. Tracking errors under the proposed controller with initial states $q_0 = q_{0ref} - \delta \mathbf{1}_6$, where $\delta \in \{0.1, 0.2, 0.3\}$ and $\mathbf{1}_6$ denotes a 6-D vector of ones. (a)–(f) Δx_1 , Δy_1 , ΔL_1 , Δx_2 , Δy_2 , and ΔL_2 .

TABLE IV

METE FOR THE PROPOSED CONTROLLER UNDER DIFFERENT INITIAL CONDITIONS IN CONVERGING AND STEADY STAGE

Initial State	METE at [0, 2]s	METE at [5, 10]s
$q_{0ref} - 0.1 \times \mathbf{1}_6$	2.0902×10^{-2}	1.6011×10^{-5}
$q_{0ref} - 0.2 \times \mathbf{1}_6$	4.1920×10^{-2}	1.6101×10^{-5}
$q_{0ref} - 0.3 \times \mathbf{1}_6$	6.6441×10^{-2}	1.6182×10^{-5}

gible overshoot. Although ARBF-FSCIS exhibits comparable convergence performance, it suffers from larger oscillations in the steady-state phase. DRNN-FTSMC demonstrates inferior accuracy, with slower convergence and higher overshoot. FWRL and FOACRL achieve moderate tracking performance, with longer settling time than the proposed method and ARBF-FSCIS during the transient phase. Their steady-state errors are lower than DRNN-FTSMC but higher than the proposed method. In addition, both controllers exhibit overshoot in the tracking error of ΔL_2 . In contrast, the proposed controller exploits the constrained control input more effectively, leading to faster convergence and lower steady-state error. These performance differences are further quantified in Table III, which summarizes the mean Euclidean tracking error (METE) of soft manipulator's configuration state over the intervals [0, 2] s and [5, 10] s for the five controllers. Compared with baseline fractional-order controllers, the proposed method reduces the METE by 23.1% in the transient phase and by 87.6% in steady state.

3) *Performance of the Proposed Controller Under Different Initial Conditions:* To validate the effectiveness of the controller under different test cases, different initial conditions are considered. All control parameters remain unchanged except for the barrier function, which is adjusted to accommodate cor-

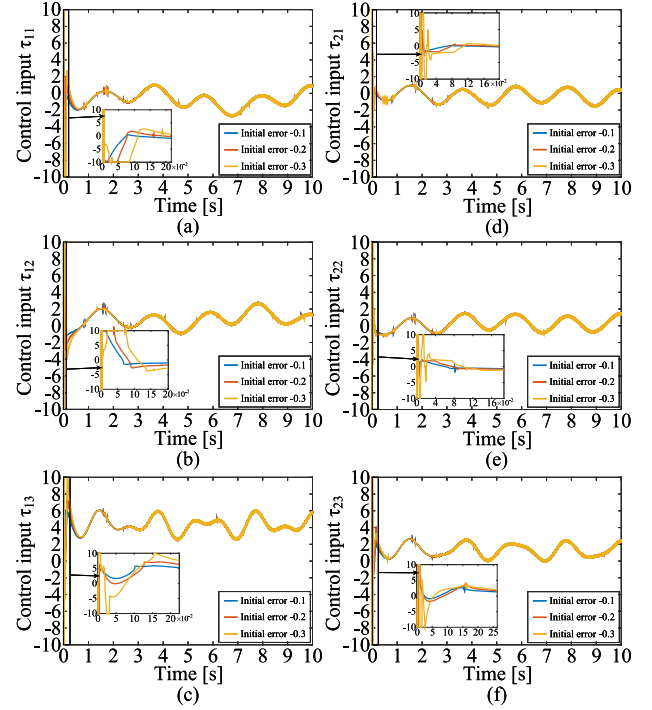


Fig. 13. Saturated control input under the proposed controller for the same initial conditions as that in Fig. 12. (a)–(f) τ_{11} , τ_{12} , τ_{13} , τ_{21} , τ_{22} , and τ_{23} .

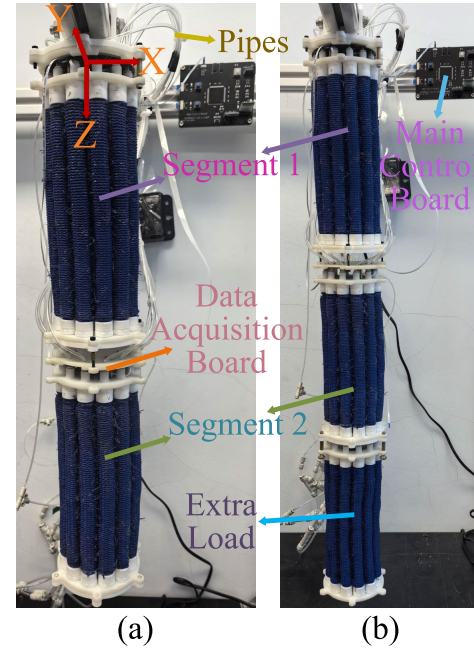


Fig. 14. Experimental setup. (a) Two-segment soft manipulator. (b) Soft manipulator with upper two actuated segments and an unactuated segment serving as external load. The learned dynamics and controllers are implemented on the main control board.

responding initial conditions. The tracking error and saturated control inputs are illustrated in Figs. 12 and 13, respectively.

As shown in Fig. 12, under all three tested initial conditions, the tracking error converges rapidly, with steady-state errors in all six configuration dimensions remaining below 3×10^{-5} . Larger initial errors result in slightly increased overshoot

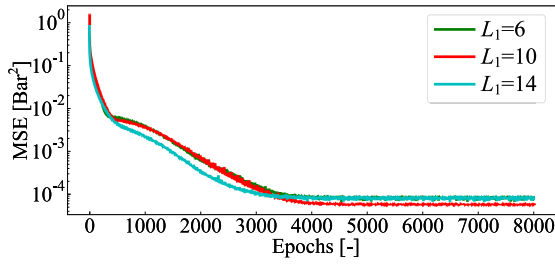


Fig. 15. MSE of predicted pressure input for fPLCS-DeLaN with $L_1 = 6, 10, 14$.

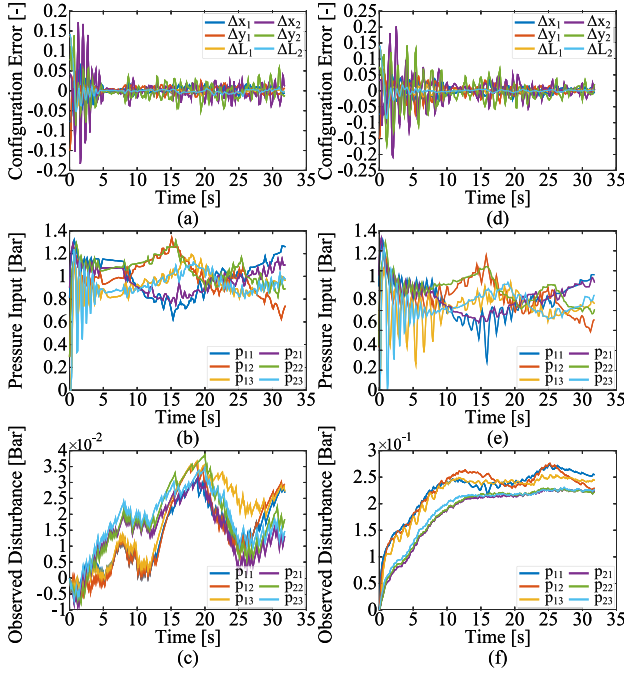


Fig. 16. Experiments results of the proposed controller on the R-shaped reference. (a)–(c) Configuration-space tracking error, chamber pressure input, and estimated lumped uncertainty under nominal conditions. (d)–(f) Corresponding results under external disturbances.

during the transient phase. Table IV reports the METE of the soft manipulator's configurations under different initial states. The METE over the interval $[0, 2]$ s increases approximately linearly with the magnitude of the initial deviation, whereas the steady-state tracking error varies by less than 1.07% across all cases. These results indicate that the proposed controller maintains both stability and robustness despite variations in initial conditions. Fig. 13 further shows that the control input approach its saturation value as the initial error increases. Nevertheless, the proposed compensation mechanism effectively mitigates the adverse effects of actuator saturation, ensuring stable and consistent performance.

V. REAL-WORLD EXPERIMENTS

A. Experimental Setup

As shown in Fig. 14(a), the two-segment soft manipulator with pipes attached for pneumatic actuation is used in experiments. Each segment is actuated by 12 pneumatic chambers, organized as three groups of four. Each group receives an

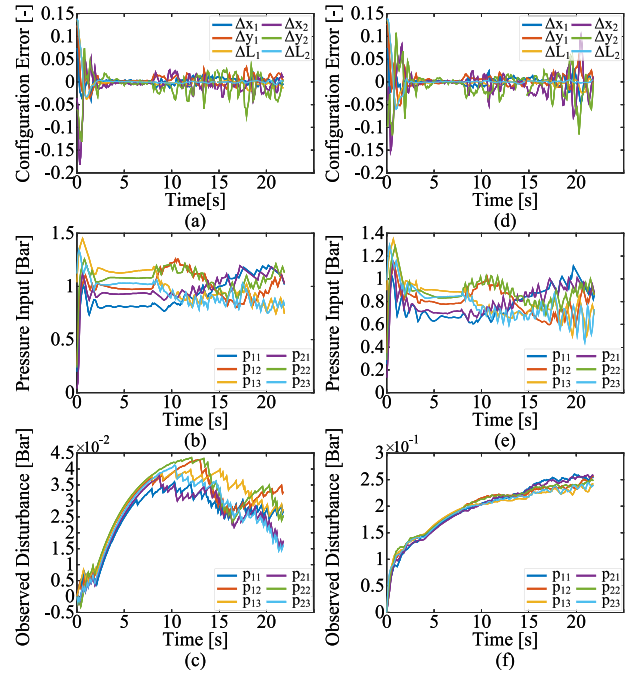


Fig. 17. Experiments results of the proposed controller on the S-shaped reference. (a)–(c) Configuration-space tracking error, chamber pressure input, and estimated lumped uncertainty under nominal conditions. (d)–(f) Corresponding results under external disturbances.

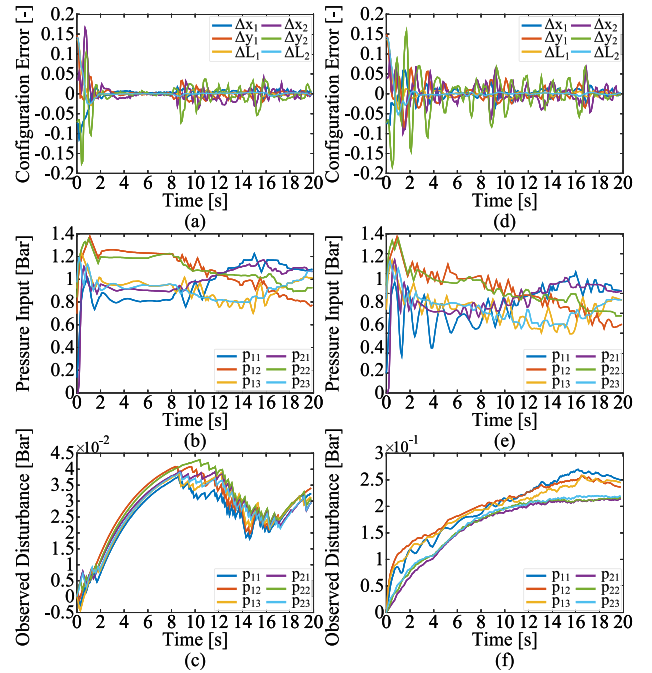


Fig. 18. Experiments results of the proposed controller on the L-shaped reference. (a)–(c) Configuration-space tracking error, chamber pressure input, and estimated lumped uncertainty under nominal conditions. (d)–(f) Corresponding results under external disturbances.

independent pressure input in $[0, 1.5]$ bar. The flexible section of each segment measures 208 mm in length and 75 mm in diameter, with chamber diameters of 15 mm. The weight of a single segment is around 0.6 kg. Configuration sensing is achieved through embedded angle and displacement sensors.

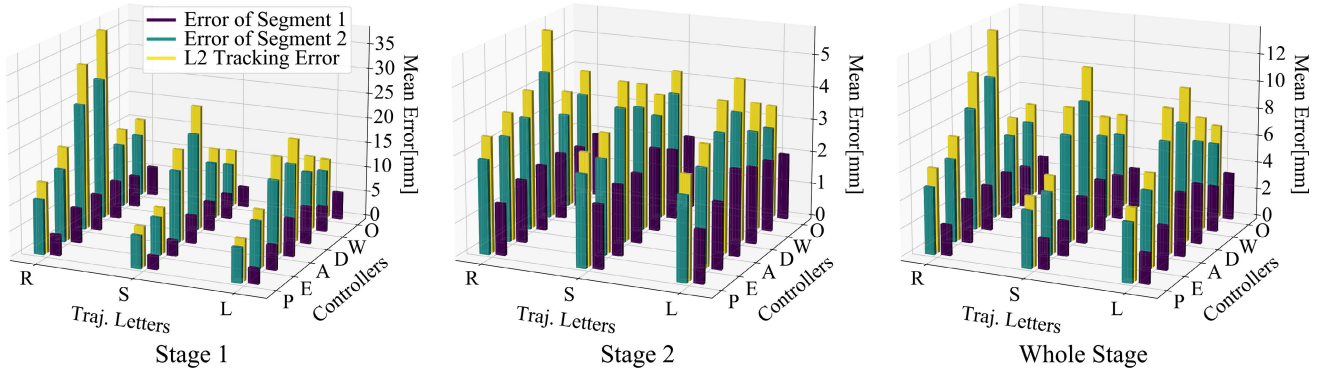


Fig. 19. METE in experiments. The letter “P,” “E,” “A,” “D,” “W,” and “O” represent the proposed controller, the proposed controller with external disturbance, ARBF-FSCIS, DRNN-FTSMC, FWRL, and FOACRL, respectively.

The end-effector’s position and orientation are computed via multijoint kinematic models [41] implemented on the onboard data acquisition system. A variable stiffness module tunes the effective Young’s modulus to 1×10^4 Pa. The mapping between configuration space and task space follows the approach in [42], while the relationship between chamber pressure and generated torque is based on the method in [43].

To verify the tracking performance, adaptability, and generalizability, we compare the proposed controller against four baselines on three letter-shaped trajectories. Each trajectory is simultaneously tracked by the end-effectors of both segments at different scales and heights within a specific time horizon. To further assess adaptability, an experimental scenario with substantial external disturbances is considered, in which an unactuated mechanical segment is mounted on the end of the soft arm [see Fig. 14(b)], thereby inducing dynamics variations. Segment 1 tracks the letters R (24 s), S (14 s), and L (12 s) on the plane $z = 235$ mm. Segment 2 tracks geometrically similar trajectories scaled by a factor of 2.5 on the plane $z = 470$ mm, with identical time durations. Prior to trajectory tracking, the controllers are also evaluated in a static control test, moving the manipulator from its initial state to the starting state of the reference trajectories within 8 s. For clarity, we define the static control, trajectory tracking, and the overall process as “Stage 1,” “Stage 2,” and “Whole stage,” respectively. All reference trajectories, nominal dynamics, and control algorithms are implemented on the main control board.

B. Data Acquisition and Model Learning via fPLCS-DeLaN

To acquire sufficient data for dynamics learning, random pressure inputs are generated within the 6-D space $([0, 1.5] \text{ bar})^6$ and applied to the gas valve cabinet at a rate of 100 Hz. The resulting configurations and their derivatives form time-series state-control pairs. For training fPLCS-DeLaN, 80 000 consecutive state-control samples are selected. Considering inevitable sensor errors, a Bayesian neural network [44] with the same architecture as fPLCS-DeLaN is employed, where the learned weights follow Gaussian distributions rather than fixed values. The Bayesian network is trained via variational inference, with the Evidence Lower Bound (ELBO) loss function formulated as follows:

$$L_{\text{baye}} = -\mathbb{E}_{r(\theta)} \ln(p(D|\theta)) + \lambda \text{KL}(r(\theta) \| p(\theta)) \quad (30)$$

where $\lambda > 0$ weights the KL regularization term. The first term denotes the learning error, the second serves as a complexity penalty. D denotes the dataset of configuration states at the next time step, θ collects all network weights. $r(\theta)$ and $p(\theta)$ are posterior distributions of variational approximation and parameter prior distribution. $\text{KL}(\cdot)$ is the Kullback–Leibler divergence function defined in (14).

For fractional-order dynamics learning, we set $L_2 = 4$ based on simulation results. We compare fPLCS-DeLaN with $L_1 = 6, 10, 14$ in terms of modeling error, defined as the mse of the predicted pressure input $E_{\text{mod}} = \text{mean}_{q,\tau}[\|\hat{\tau} - \tau\|^2]$. The hyperparameters are detailed in Table S2 of the Supplementary Materials. Fig. 15 plots the modeling error with training epochs. Among the tested configurations, fPLCS-DeLaN with $L_1 = 10$ yields the lowest final modeling error. In the rest of this article, $L_1 = 10$ is chosen.

C. Experimental Results for the Proposed Controller

Trajectory tracking experiments were conducted on three letter-shaped trajectories (R, S, L) for the **P**roposed controller, the proposed controller with **E**xternal load, **AR**BF-FSCIS, **DR**NN-FTSMC, **FW**RL, and **FO**ACRL. The hyperparameters and time consumption of these controllers are detailed in Table S4 of the Supplementary Materials. To ensure fair comparison while balancing control performance and computational overhead, all controllers run at a frequency of 20 Hz.

Figs. 16–18 present the configuration-space tracking errors, actuation pressures, and estimated uncertainty for the proposed controller, both in nominal conditions and under external load, across the three tasks. The 3-D bar chart of METE is presented in Fig. 19, summarizing controller performance across different tracking tasks and control stages. The proposed controller consistently outperforms all four baselines across all tested trajectories, achieving a 7.37%, 32.91%, and 24.61% reduction in L2 tracking error compared to the best-performing baseline on trajectories “R,” “S,” and “L,” respectively. Even under substantial external load, the proposed controller demonstrates superior robustness and performance. It outperforms all baselines on “S” and “L” by more than 22.6% and 33.1%, respectively. It also performs comparably to the top-performing baselines (FWRL and FOACRL) on trajectory “R,” with less than a 15% increase in mean tracking error of

the whole stage. This verifies the robustness of the proposed controller considering the magnitude of external load. The trajectory tracking results for all three reference under various controllers and conditions, with a heatmap visualization of tracking error included to highlight performance differences, are further illustrated in Fig. S2 of Supplementary Materials.

VI. CONCLUSION

This work addresses the challenge of modeling and control for systems with memory effects by proposing a unified framework. It composes fPLCS-DeLaN for model learning, T2F-CRNN for observing lumped uncertainties and a fully fractional-order controller. Validated on a soft manipulator, fPLCS-DeLaN achieves modeling errors one to two orders of magnitude lower than InLaN and SAE-DeLaN, with less than a 15% increase in computational time. Integrating with the T2F-CRNN observer, the proposed fractional-order controller reduces transient and steady-state tracking errors by 23.1% and 87.6%, respectively. Real-world experiments further confirm its superior performance and applicability under practical constraints such as actuator saturation and model mismatch.

To the best of our knowledge, this is among the first efforts to extend DeLaN to fractional-order dynamics modeling. The co-design of a learned dynamics model and a practical finite-time stable fractional-order controller under practical constraints remains an open challenge, and this work provides a feasible solution. Future research includes integrating high-fidelity dynamic models such as Cosserat rod models, as well as more complex physical phenomena—like hysteresis and self-excited vibrations—into physics-informed learning, and developing fault-tolerant fractional-order controllers that explicitly deal with sensor, actuator faults and time delay.

REFERENCES

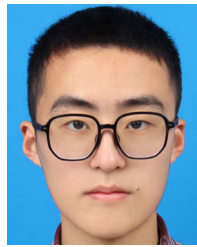
- [1] Y. Kao, Y. Cao, and Y. Chen, "Projective synchronization for uncertain fractional reaction-diffusion systems via adaptive sliding mode control based on finite-time scheme," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 15638–15646, Nov. 2024.
- [2] Y. Yan, H. Zhang, J. Sun, and Y. Wang, "Sliding mode control based on reinforcement learning for T-S fuzzy fractional-order multiagent system with time-varying delays," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 8, pp. 10368–10379, Aug. 2024.
- [3] Z. Wang and G. Li, "An inverse problem of determining the fractional order in a fractional SVIRS epidemic model based on the asymptotic solution," *J. Inverse Ill-posed Problems*, vol. 33, no. 6, pp. 813–827, Dec. 2025.
- [4] A. Djeddi, A. T. Azar, A. Djari, D. Dib, S. Alshathri, and W. El-Shafai, "Robust fault diagnosis of fractional order Takagi–Sugeno systems with uncertainties in premise variables," *Sci. Rep.*, vol. 15, no. 1, p. 32163, Sep. 2025.
- [5] A. T. Azar et al., "Fractional-order Euler–Lagrange dynamic formulation and control of asynchronous switched robotic systems," in *Proc. 3rd Int. Conf. Sustain. Computing*. Cham, Switzerland: Springer, 2022, pp. 479–490.
- [6] M. K. Mishra, A. K. Samantaray, and G. Chakraborty, "Fractional-order bouc-wen hysteresis model for pneumatically actuated continuum manipulator," *Mech. Mach. Theory*, vol. 173, Mar. 2022, Art. no. 104841.
- [7] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *J. Comput. Phys.*, vol. 378, pp. 686–707, Feb. 2019.
- [8] F. Wang et al., "Inherently interpretable physics-informed neural network for battery modeling and prognosis," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 1, pp. 1145–1159, Nov. 2023.
- [9] C. Oszkinat, S. E. Luczak, and I. G. Rosen, "Uncertainty quantification in estimating blood alcohol concentration from transdermal alcohol level with physics-informed neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 8094–8101, Oct. 2023.
- [10] Z. Fang, "A high-efficient hybrid physics-informed neural networks based on convolutional neural network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5514–5526, Oct. 2022.
- [11] T. Kapoor, A. Chandra, D. M. Tartakovsky, H. Wang, A. Núñez, and R. Dollevoet, "Neural oscillators for generalization of physics-informed machine learning," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 13059–13067.
- [12] B. Hillebrecht and B. Unger, "Rigorous a posteriori error bounds for PDE-defined PINNs," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 1, pp. 1583–1593, Jan. 2025.
- [13] G. Pang, L. Lu, and G. E. Karniadakis, "FPINNs: Fractional physics-informed neural networks," *SIAM J. Scientific Comput.*, vol. 41, no. 4, pp. A2603–A2626, Jan. 2019.
- [14] Z. Hu, Z. Zhang, G. E. Karniadakis, and K. Kawaguchi, "Score-fPINN: Fractional score-based physics-informed neural networks for high-dimensional Fokker-Planck-Levy equations," 2024, *arXiv:2406.11676*.
- [15] Z. Hu, K. Kawaguchi, Z. Zhang, and G. E. Karniadakis, "Tackling the curse of dimensionality in fractional and tempered fractional PDEs with physics-informed neural networks," 2024, *arXiv:2406.11708*.
- [16] J. Zhang, X. Tian, and C. Zhou, "FFNN: Fractional order basis function multi-step neural network method for fractional partial differential equations," *Appl. Soft Comput.*, vol. 185, Dec. 2025, Art. no. 113907.
- [17] M. Lutter, C. Ritter, and J. Peters, "Deep Lagrangian networks: Using physics as model prior for deep learning," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023, pp. 1–17.
- [18] B. Johns, Z. Zhou, and E. Abdi, "ServoLNN: Lagrangian neural networks driven by servomechanisms," 2025, *arXiv:2502.19802*.
- [19] X. Shao, L. Xu, G. Sun, W. Yao, L. Wu, and C. Della Santina, "Self-attention enhanced dynamics learning and adaptive fractional-order control for continuum soft robots with system uncertainties," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 18694–18708, 2025.
- [20] X. Shao, G. Sun, W. Yao, X. Li, and O. Zhang, "Fractional-order resolved acceleration control for free-floating space manipulator with system uncertainty," *Aerosp. Sci. Technol.*, vol. 118, Nov. 2021, Art. no. 107041.
- [21] X. Shao et al., "Model-based control for soft robots with system uncertainties and input saturation," *IEEE Trans. Ind. Electron.*, vol. 71, no. 7, pp. 7435–7444, Jul. 2024.
- [22] Y. Chu, J. Fei, and S. Hou, "Adaptive global sliding-mode control for dynamic systems using double hidden layer recurrent neural network structure," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 4, pp. 1297–1309, Apr. 2020.
- [23] S. Tian and T. Zhao, "Self-organizing interval type-2 function-link fuzzy neural network control for uncertain manipulators under saturation: A predefined-time sliding-mode approach," *Appl. Soft Comput.*, vol. 165, Nov. 2024, Art. no. 112064.
- [24] L.-J. Li et al., "Self-organizing type-2 fuzzy double loop recurrent neural network for uncertain nonlinear system control," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 6451–6465, Apr. 2025.
- [25] B. Abdelhamid and C. Mohamed, "Robust fuzzy adaptive fault-tolerant control for a class of second-order nonlinear systems," *Int. J. Adapt. Control Signal Process.*, vol. 39, no. 1, pp. 15–30, Jan. 2025.
- [26] A. Bounemeur and M. Chemachema, "Adaptive fuzzy fault-tolerant control using nussbaum-type function with state-dependent actuator failures," *Neural Comput. Appl.*, vol. 33, no. 1, pp. 191–208, Jan. 2021.
- [27] A. Bounemeur and M. Chemachema, "General fuzzy adaptive fault-tolerant control based on nussbaum-type function with additive and multiplicative sensor and state-dependent actuator faults," *Fuzzy Sets Syst.*, vol. 468, Jun. 2023, Art. no. 108616.
- [28] B. Jin, *Fractional Differential Equations*. Cham, Switzerland: Springer, 2021.
- [29] G. Sun, L. Wu, Z. Kuang, Z. Ma, and J. Liu, "Practical tracking control of linear motor via fractional-order sliding mode," *Automatica*, vol. 94, pp. 221–235, Aug. 2018.
- [30] S. Kumari, R. K. Pandey, and R. P. Agarwal, "High-order approximation to generalized caputo derivatives and generalized fractional advection–diffusion equations," *Mathematics*, vol. 11, no. 5, p. 1200, Feb. 2023.

- [31] H. Zhou et al., "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 12, pp. 11106–11115.
- [32] Y. Wei, X. Shao, J. Liu, S. Zhang, and L. Xu, "Dynamics modeling of soft robots based on attention-enhanced Lagrangian deep neural networks," in *Proc. 3rd Conf. Fully Actuated Syst. Theory Appl. (FASTA)*, May 2024, pp. 200–207.
- [33] F. Chao, D. Zhou, C.-M. Lin, L. Yang, C. Zhou, and C. Shang, "Type-2 fuzzy hybrid controller network for robotic systems," *IEEE Trans. Cybern.*, vol. 50, no. 8, pp. 3778–3792, Aug. 2020.
- [34] S. C. Yogi, V. K. Tripathi, and L. Behera, "Adaptive integral sliding mode control using fully connected recurrent neural network for position and attitude control of quadrotor," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 12, pp. 5595–5609, Dec. 2021.
- [35] C.-M. Lin, M.-S. Yang, F. Chao, X.-M. Hu, and J. Zhang, "Adaptive filter design using type-2 fuzzy cerebellar model articulation controller," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 10, pp. 2084–2094, Oct. 2016.
- [36] L. Chen, S.-L. Dai, and C. Dong, "Adaptive optimal tracking control of an underactuated surface vessel using actor-critic reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 6, pp. 7520–7533, Jun. 2022.
- [37] C. Wang, L. Cui, M. Liang, J. Li, and Y. Wang, "Adaptive neural network control for a class of fractional-order nonstrict-feedback nonlinear systems with full-state constraints and input saturation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6677–6689, Nov. 2022.
- [38] Y.-L. Wang, H. Jahanshahi, S. Bekiros, F. Bezzina, Y. Chu, and A. A. Aly, "Deep recurrent neural networks with finite-time terminal sliding mode control for a chaotic fractional-order financial system with market confidence," *Chaos, Solitons & Fractals*, vol. 146, Jun. 2023, Art. no. 110881.
- [39] D. Li and J. Dong, "Fuzzy weight-based reinforcement learning for event-triggered optimal backstepping control of fractional-order nonlinear systems," *IEEE Trans. Fuzzy Syst.*, vol. 32, no. 1, pp. 214–225, Jan. 2024.
- [40] D. Li and J. Dong, "Fractional-order systems optimal control via actor-critic reinforcement learning and its validation for chaotic MFET," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 1173–1182, 2025.
- [41] X. Shao, G. Sun, C. Xue, and X. Li, "Nonsingular terminal sliding mode control for free-floating space manipulator with disturbance," *Acta Astronautica*, vol. 181, pp. 396–404, Jan. 2021.
- [42] C. Della Santina, A. Bicchi, and D. Rus, "On an improved state parametrization for soft robots with piecewise constant curvature and its use in model based control," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 1001–1008, Apr. 2020.
- [43] C. D. Santina, A. Bicchi, and D. Rus, "Dynamic control of soft robots with internal constraints in the presence of obstacles," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Nov. 2019, pp. 6622–6629.
- [44] L. Menicali, D. H. Richter, and S. Castruccio, "Bayesian neural networks with physics-informed priors with application to boundary layer velocity," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 18048–18061, Jun. 2025.



Xiangyu Shao (Member, IEEE) received the B.S. degree in automation from Harbin Engineering University, Harbin, China, in 2016, and the M.S. and Ph.D. degrees in control science and engineering from Harbin Institute of Technology, Harbin, in 2018 and 2022, respectively.

He is currently an Associate Professor with the School of Astronautics, Harbin Institute of Technology. His research interests include space robots, soft robots, sliding mode control, and fractional order control.



Linke Xu (Graduate Student Member, IEEE) received the B.S. degree in automation from Harbin Institute of Technology, Harbin, China, in 2024, where he is currently pursuing the Ph.D. degree in control science and engineering.

His research interests include soft robots, sliding mode control, fractional-order control, and learning-based dynamics modeling.



Shaojie Zhang received the B.S. degree in automation from Chongqing University, Chongqing, China, in 2023. He is currently pursuing the Ph.D. degree in control science and engineering with the School of Astronautics, Harbin Institute of Technology, Harbin, China.

His research interests include dynamics identification, nonlinear optimization, and legged robots.



Guanghui Sun (Senior Member, IEEE) received the B.S. degree in automation and the M.S. and Ph.D. degrees in control science and engineering from Harbin Institute of Technology, Harbin, China, in 2005, 2007, and 2010, respectively.

He is currently a Professor with the Department of Control Science and Engineering, Harbin Institute of Technology. His research interests include fractional-order systems, networked control systems, and sliding mode control.



Chengwei Wu received the B.S. degree from the Arts and Science College, Bohai University, Jinzhou, China, in 2013, the M.S. degree from Bohai University, in 2016, and the Ph.D. degree from Harbin Institute of Technology, Harbin, China, 2021.

From July 2015 to December 2015, he was a Research Assistant with the Department of Mechanical Engineering, The Hong Kong Polytechnic University, Hong Kong. From 2019 to 2021, he was a Joint-Ph.D. Student at the Department of Cognitive Robotics, Delft University of Technology, Delft, The Netherlands. He is currently an Associate Professor with Harbin Institute of Technology. His research interests include reinforcement learning, sliding mode control, and cyber-physical systems.